

UMF — Bounded-Universe Topology

A literature-review-and-positioning analysis under the reflective-boundary commitment
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Preamble and scope

This document is a literature-review-and-positioning analysis of the bounded-universe topology question under the Unified Metric Framework. It is the first UMF-specific empirical engagement document produced under the framework's standing rules, and it is undertaken without specialist input as a candidate first analysis identified in the Empirical Engagement document v1.2.

The analysis is at the structural-inference level rather than the quantitative-constraint level. It does not claim numerical support for the framework, does not derive equations specific to UMF's commitments, and does not assert that UMF predictions match observations at any specific significance level. What it does is survey the relevant published literature (matched-circles methodology and Planck topology bounds, Israel junction condition apparatus and contemporary thin-shell applications), establish that UMF's bounded-universe commitment is not in tension with the published Planck topology results at the structural-inference level, and identify clear targets for the specialist engagement.

This scope is deliberately bounded. The literature review revealed that the methodology adaptation required to convert the existing matched-circles analysis into a quantitative test of UMF's reflective-boundary commitment is substantial and properly specialist work. Producing a quantitative constraint without that adaptation would be the kind of overreach that the framework's standing rules are designed to prevent. The document's value is in establishing literature context and clear specialist targets rather than in claiming quantitative results.

What UMF commits to that this analysis tests

The Unified Metric Framework v2.2 commits to two specific structural positions that the bounded-universe topology analysis engages.

First, the framework commits to a bounded universe (Region A) with finite spatial extent, separated from a primordial state (Region C) by an active wavefront boundary (Region B). The bounded-universe commitment is foundational; it is Postulate 2 of the foundational logic and is structurally load-bearing for the framework's subsequent commitments.

Second, the framework commits to the wavefront's inner surface being totally reflective to all properties of established spacetime — electromagnetic radiation, gravitational waves, matter, particles. The only quantity that crosses the boundary is gravitational tension T_G itself, in the direction of relaxation toward primordial equilibrium. This is Postulate 3 of the foundational logic v1.2 and is operationalised in the formal framework v2.2 as a specific commitment about the wavefront's physical properties.

These two commitments together produce a specific empirical implication: if Region A is small enough that CMB photons can make round-trip reflections off the wavefront's inner surface within the observable redshift range, the temperature pattern of the cosmic microwave background should carry signatures of those

reflections at the largest angular scales. This is the empirical implication that the matched-circles methodology in the multiply-connected-topology literature was developed to detect, applied to a structurally different geometry.

The matched-circles methodology and Planck topology bounds

The matched-circles methodology for detecting non-trivial cosmic topology was developed by Cornish, Spergel, and Starkman in "Circles in the Sky: Finding Topology with the Microwave Background Radiation" (Class. Quant. Grav. 15, 2657-2670, 1998; arXiv:astro-ph/9801212). The core insight of the methodology is geometric: if the universe is finite and the diameter of the universe is smaller than the distance to the surface of last scattering, then a single emission patch at last scattering can be observed via multiple light paths, producing matched temperature patterns along circles in the CMB sky. The intersection of multiple last-scattering-surface copies (as seen by the observer through different paths) produces circular intersections in the sky regardless of the specific topology or background geometry, which is why the methodology is generic across topology classes.

Scope of validity

The methodology was developed and is rigorously valid in the context of multiply-connected topologies of FLRW cosmologies. In this context, the "multiple light paths" arise from topological identification — the same physical point in space appears at multiple locations in the universal cover. The light reaching the observer from the same emission patch via different paths is geometrically the same emission patch seen through the topological identifications.

The methodology does not directly apply to reflective-boundary cosmologies where the multiple light paths arise from physical reflection off a boundary rather than from topological identification. The matched-circles signature would be expected by analogy (multiple paths to the same emission patch produce correlated patches in the CMB sky regardless of whether the paths arise from identification or from reflection), but the specific predictions for circle radii, distributions, and statistical properties would differ between the topology case and the reflective-boundary case. Methodology adaptation is required to make the analysis quantitative for the reflective-boundary case.

Published Planck bounds

The Planck collaboration applied the matched-circles methodology to their CMB temperature data and reported bounds in several publications.

Planck 2013 Results XXVI (Background geometry and topology of the Universe; Astron. Astrophys. 571, A26, 2014; arXiv:1303.5086) performed the first application of the matched-circles search to Planck data and reported no detection of matched circles, with the analysis extending the lower bound on fundamental-domain size relative to earlier WMAP results.

Planck 2015 Results XVIII (Background geometry and topology of the Universe; arXiv:1502.01593) extended the analysis with both matched-circles searches and likelihood-based methods for specific topologies. For the cubic torus (T3) topology, the analysis bounded the radius of the largest sphere inscribed in the fundamental domain at $R_i > 0.97 \times \chi_{\text{rec}}$ at 99% confidence level from the matched-circles search using polarisation data, where χ_{rec} is the comoving distance to the recombination surface. For the slab (T1) topology, the bound is $R_i > 0.56 \times \chi_{\text{rec}}$.

The WMAP-based extension by Vaudrevange, Starkman, Cornish, and Spergel (Phys. Rev. D 86, 083526, 2012; arXiv:1206.2939) reported a more refined bound: the injectivity radius at our location is at least 98.5% of the diameter of the last scattering surface, approximately 26 Gpc. The Bielewicz, Banday, and Górski analysis (MNRAS 421, 1064, 2012; arXiv:1303.4004) reported a lower bound on the size of the fundamental domain of approximately 27.9 Gpc for a flat universe.

These bounds are pushing right up against the limit of what the matched-circles methodology can detect — the observable universe diameter is approximately 28.3 Gpc, so the bounds essentially say that no matched circles have been detected at any size the methodology is sensitive to.

What the absence of matched circles signifies

In the multiply-connected-topology context, the absence of matched circles places a lower bound on the size of the fundamental polyhedral domain. Topologies with fundamental domains smaller than approximately the observable universe diameter would have produced detectable matched circles and are therefore excluded at the confidence levels reported. Topologies with fundamental domains larger than this scale produce signatures (if any) below the methodology's detection threshold.

In the reflective-boundary context (UMF), the analogous signature would be CMB photons reaching the observer both directly from the surface of last scattering and after one or more reflections off the wavefront's inner surface. The matched-circles methodology, with appropriate adaptation, would detect signatures of round-trip reflection paths if the wavefront is close enough to the observer for such paths to fall within the observable redshift range. The absence of matched circles in Planck data is, at the structural-inference level, consistent with the wavefront being far enough from the observer that round-trip reflection paths do not fall within the observable redshift range, which corresponds to Region A being at least approximately the diameter of the observable universe.

This is a structural inference, not a quantitative constraint. The specific bound on Region A's size under UMF's commitments would require the methodology adaptation that has not been performed in the published literature for reflective-boundary cosmologies of UMF's specific type. What can be stated honestly is that UMF's bounded-universe commitment is not in tension with the existing Planck topology results, with the empirical situation properly described as "no matched-circles signatures observed at the methodology's detection threshold, consistent with Region A being larger than the observable universe diameter under UMF's structural commitments."

The Israel junction condition apparatus and contemporary applications

The Israel junction conditions provide the mathematical framework for treating a thin hypersurface separating two regions of spacetime with potentially different metrics. The framework is the appropriate apparatus for UMF's wavefront and is identified as such in both prior cold reviews (ChatGPT, Claude Opus 4.7 incognito) as a candidate import for the framework's formal scaffold.

Foundational papers

Israel's "Singular hypersurfaces and thin shells in general relativity" (Nuovo Cimento B 44, 1, 1966) is the foundational paper. The Israel junction conditions are two: the induced metric on the shell must be continuous across it (the spacetime is glued

together smoothly), and the jump in extrinsic curvature across the shell is related to the shell's surface stress-energy through the equation $[K_{ij}] - h_{ij} [K] = -8\pi G S_{ij}$, where the bracket denotes the discontinuity across the shell, h_{ij} is the induced metric, and S_{ij} is the shell's surface stress-energy tensor. The methodology has been continuously developed since its publication and is standard apparatus in classical general relativity.

Berezin, Kuzmin, and Tkachev "Dynamics of bubbles in general relativity" (Phys. Rev. D 36, 2919, 1987) extended the Israel formalism to dynamical thin shells, working out the equations of motion for shells with various surface stress-energy contents and producing the modified Friedmann-like equation for the shell's radius evolution. The key result is that a shell with constant surface tension produces a term in the modified Friedmann equation that behaves as a cosmological constant in the late-time limit. This is the structural origin of dark-energy-like behaviour in thin-shell cosmologies.

Contemporary peer-reviewed applications

The Israel-Berezin formalism has been applied in many contexts since its development. Two recent applications that are structurally relevant to UMF's commitments are worth flagging.

Schubert's "Timelike Thin-Shell Evolution in Gravitational Collapse: Classical Dynamics and Thermodynamic Interpretation" (Entropy 28, 96, 2026; published January 2026) uses Israel junction conditions to separate a regular de Sitter interior from a Schwarzschild or Schwarzschild–de Sitter exterior. The configuration is structurally close to UMF's setup in that a thin timelike hypersurface separates two distinct geometric regions with the Israel junction conditions providing the matching. The paper is peer-reviewed and demonstrates that the Israel apparatus continues to produce publishable results when applied to configurations similar to UMF's wavefront. Specific results include identification of a deceleration mechanism at a balance radius, classification of the allowable radial domain for outward evolution, and bounded curvature invariants throughout the shell-supported spacetime region.

The broader brane cosmology literature (Randall-Sundrum 1999 and successors, the Israel-junction-condition methodology developed in that programme) provides additional context. The brane cosmology setups use timelike thin shells in 5-dimensional ambient spacetime, which is geometrically different from UMF's 4-dimensional setup but uses the same junction-condition apparatus. The methodological development in brane cosmology is therefore relevant to UMF as a demonstration of how the apparatus has been extensively developed and tested.

What the apparatus provides for UMF

The Israel junction condition apparatus provides UMF with the mathematical framework for treating the wavefront as a thin hypersurface with surface stress-energy, separating Region A (interior FRW-like cosmology) from Region C (primordial equilibrium). The apparatus produces the modified Friedmann-like equation for the wavefront's evolution as a function of its surface stress-energy. Under UMF's commitments, the wavefront's surface stress-energy is related to the cosmological surface tension $P_P \times I_P$ at maximum sustainable level, with the cosmic lifecycle phases corresponding to different effective equations of state for the wavefront's surface tension.

The apparatus does not by itself produce UMF-specific predictions. UMF's commitments need to be made specific (the wavefront's surface stress-energy tensor in UMF terms, the equation of state of the wavefront during plateau and relaxation

phases, the relationship between the framework's gravitational tension T_G and the wavefront's surface properties) before the apparatus can be applied to produce predictions. This is the specialist work the framework's open theoretical questions identify.

Structural inference and clear specialist targets

The structural inference produced by this analysis is the following. Under UMF's commitments to a bounded universe with totally reflective wavefront, the matched-circles methodology adapted from the multiply-connected-topology literature would detect signatures of round-trip CMB reflection paths if Region A is small enough for such paths to fall within the observable redshift range. The absence of matched-circles signatures in Planck data, at the detection threshold reported by the Planck collaboration and the WMAP-based extensions, is consistent with Region A being at least approximately the diameter of the observable universe under UMF's commitments. The framework's bounded-universe commitment is therefore not in tension with the existing observational results.

This structural inference is at the level the framework's standing rules permit without specialist input. It does not claim a quantitative bound on Region A's size, because such a bound would require methodology adaptation that has not been performed in the literature for UMF's specific reflective-boundary geometry. It does not claim that UMF predicts the observed CMB temperature pattern, because such a prediction would require specifying the wavefront's formal stress-energy content under UMF's commitments. It claims only that the existing observational situation is consistent with the framework's structural commitments.

Specialist work items identified

The bounded-universe topology analysis identifies specific work items for the specialist engagement, each of which would convert the structural inference into a quantitative result if completed.

- **Methodology adaptation.** Derive the matched-circles signature analogue for reflective-boundary geometry. This requires working out how a CMB photon emitted at the surface of last scattering, propagating to the wavefront, reflecting off the wavefront's inner surface, and returning to the observer, produces correlated patches in the CMB sky. The geometric calculation should yield predictions for the radii, locations, and statistical properties of the resulting matched circles as a function of Region A's size and the wavefront's reflective properties.
- **Wavefront stress-energy specification.** Specify the surface stress-energy tensor S_{ij} of the wavefront under UMF's commitments. This includes the surface tension at the cosmological surface tension scale, the relationship between the wavefront's surface properties and the framework's gravitational tension T_G , and the equation of state of the wavefront during the plateau and relaxation phases. The Israel junction conditions then produce the modified Friedmann-like equation for Region A under UMF's specific commitments.
- **Quantitative constraint derivation.** Apply the adapted matched-circles methodology to the existing Planck data with UMF's wavefront stress-energy commitments, producing a quantitative lower bound on Region A's size under UMF. This converts the structural inference into a specific number that can be compared to other empirical constraints on the framework.
- **Birkhoff theorem reconciliation.** The reflective-boundary commitment produces signatures that are structurally distinct from standard-GR gravitational waves. The relationship between the framework's reflective boundary and the Birkhoff theorem

requires careful examination — the framework's commitment to stochastic content inside Region A partially defuses the Birkhoff issue (statistical isotropy rather than dynamical spherical symmetry), but the formal apparatus for the wavefront's reflective dynamics may still require scalar-tensor reformulation or beyond-GR commitment. This is connected to the action-principle gating issue from the v2.2 open theoretical questions.

- **Empirical engagement at correlation-class significance.** Once the methodology adaptation and stress-energy specification are complete, the analysis can produce constraint-class or correlation-class results from the existing Planck data. The pre-registration discipline established in the UMDT framework lineage applies: the predictions should be locked before re-analysing the Planck data under UMF's specific commitments, and the analysis should report at the framework's rigour standard (5σ proof, 4σ correlation, 2σ - 3σ notable only).

What this document does and does not claim

For the avoidance of doubt, and to maintain the framework's standing rule on honesty in evaluation, this document explicitly distinguishes what it claims from what it does not claim.

What this document claims

- UMF v2.2 commits to a bounded universe (Region A) with a totally reflective inner wavefront boundary, and these commitments are documented in the foundational logic v1.2 (Postulates 2 and 3) and the formal framework v2.2.
- The matched-circles methodology developed by Cornish, Spergel, and Starkman (1998) and applied to Planck data (Planck 2013 XXVI, 2015 XVIII, 2018 VII) is the relevant published methodology for testing bounded-universe topology, with its specific scope of validity in multiply-connected-topology FLRW cosmologies.
- The published Planck bounds report no detection of matched circles, consistent with topological scale below the observable universe diameter (approximately 28.3 Gpc) being excluded for the topology classes the methodology can detect.
- At the structural-inference level, UMF's bounded-universe commitment is consistent with these observational results.
- The Israel junction condition apparatus (Israel 1966, Berezin-Kuzmin-Tkachev 1987) is the relevant formal apparatus for UMF's wavefront, with contemporary peer-reviewed applications (Schubert 2026 and others) demonstrating that the apparatus is mature and produces publishable results on structurally similar configurations.
- The conversion of the structural inference into a quantitative constraint requires specialist work, with the work items identified explicitly in this document.

What this document does not claim

- It does not claim that UMF is supported by Planck topology data at any specific significance level.
- It does not claim that UMF predicts the observed CMB temperature pattern in any specific way.
- It does not claim that the matched-circles methodology has been adapted to UMF's reflective-boundary geometry in the published literature.
- It does not claim a quantitative bound on Region A's size under UMF's commitments.
- It does not derive equations specific to UMF's formal scaffold or apply the Israel junction conditions to UMF's specific wavefront stress-energy content.

- It does not produce any figures, fits, or numerical comparisons between UMF predictions and observations.

Methodology and provenance

This document is the product of a structured literature review undertaken in collaboration with Claude (Anthropic), conducted on 18 May 2026 (AEDT). The literature review proceeded by web search of the published literature on the relevant topics (cosmic topology methodology, Planck topology bounds, Israel junction conditions, contemporary thin-shell applications), with the findings evaluated under the framework's standing rules for full evaluation in UMF context before adoption.

The Holmes "Resonant-Shell Cosmology" paper (Zenodo, 2025) was encountered during the literature search and was specifically evaluated in detail (see UMF working notes, 18 May 2026). The structural commitment in that paper to a reflective-shell cosmology with Israel junction conditions confirmed the relevance of the Israel apparatus to UMF's setup. The specific numerical claims in the Holmes paper did not survive UMF's rigour standard (the recombination radius "correction" is conceptually unclear, the cavity-eigenmode CMB peak picture is structurally different from standard CMB physics in undocumented ways, the Boltzmann-code fit claims are presented without supporting code or methodology). The structural confirmation that Israel junction conditions are applied in the literature to reflective-boundary cosmologies is incorporated; the specific implementation in the Holmes paper is not.

The document's scope was deliberately bounded to the literature-review-and-positioning level after the review revealed that the methodology adaptation required for a quantitative UMF-specific analysis is properly specialist work. Overreaching to a quantitative constraint-class analysis without the methodology adaptation would have been inconsistent with the framework's standing rules. The bounded scope is intended to strengthen the framework's position for the specialist engagement by establishing literature context and identifying clear specialist targets, not to substitute for the specialist work.

Closing

The bounded-universe topology analysis at the structural-inference level establishes that UMF's bounded-universe and reflective-boundary commitments are not in tension with the existing Planck topology results. The framework can carry this finding into the specialist engagement as supporting context: the existing observational situation is consistent with the framework's structural commitments, with the work items for converting structural consistency into quantitative engagement clearly identified.

The specialist work items identified in this document are bounded and well-defined: methodology adaptation for reflective-boundary geometry, wavefront stress-energy specification under UMF's commitments, quantitative constraint derivation from the existing Planck data, Birkhoff theorem reconciliation, and pre-registered empirical engagement at correlation-class significance. Each work item is connected to the framework's open theoretical questions documented in the formal framework v2.2 (especially the action-principle gating issue, the variable-coupling Maxwell sector work, and the formal apparatus for dark matter as gravitoelectric field contribution).

This analysis is the first UMF-specific empirical engagement document produced under the framework's standing rules without specialist input. It does not claim to be

more than it is, and it does not stand alone — it is companion content to the UMF v2.2 documentation package and should be read alongside the formal framework, foundational logic, and empirical engagement documents.